

Manual

KPI Monitoring / OEE (MOC) MDE-KMO 8.2

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Contents

1	Overview of KPI Monitor / OEE.....	4
2	Efficiency report	6
3	Performance profile.....	13
4	OEE report	20

1 Overview of KPI Monitor / OEE

Purpose

Use the function package *KPI monitor/OEE* to evaluate the data collected in the MDE module (quantities, times) over different periods and relating to defined key figures.

Implementation notes

Use the function package if you want to:

- monitor and assess the performance of specific machines, machine groups or departments using KPIs.
- monitor and assess the performance of specific machines, machine groups or departments using the "Overall Equipment Effectiveness" (OEE).

Integration

The function uses the times and quantities collected in BDE or MDE to display the order progress.

Features

- Efficiency report
 - Efficiency report on quantities and times for all machines over specific periods and shifts
 - Distribution graph showing the rate of capacity utilization (utilization efficiency), scrap rate, technical efficiency and assignment utilization rate (allocation efficiency) for several machines
- Performance profile
 - Evaluations on occupancy time, machine work time, assignment utilization rate (allocation efficiency) and technical efficiency
- OEE report
 - Tabular evaluation of the "Overall Equipment Effectiveness" (OEE) to analyze the key performance indicators productivity, quality and effectiveness for all machines. Graphical display of the OEE and its components.
 - Distribution graph showing the OEE and its components for multiple machines.
 - Tabular or graphical displays using bar or pie charts.
 - Views of the top ten key performance indicators
- OEE profile
 - Tabular and shift-based evaluation of the "Overall Equipment Effectiveness" (OEE) to analyze the key performance indicators productivity, quality and effectiveness for all machines. Graphical display of the OEE and its components.

- Tabular or graphical displays using bar charts.

2 Efficiency report

Overview

Menu	Production facility management → Key performance indicators → Efficiency report
Transaction code	effrp
Function authorization	effrp

The report includes workplace/machine-related performance data for a specific period of time and a specific number of workplaces. The result depends on the selections made and therefore on the selection criteria available in the selection panel. The performances regarding quantities and durations are displayed in a graphic. The production controller gets a quick and clear overview of the performances.

Selection criteria

The application provides the following selection criteria:

Workplace

This selection criterion refers to the workplace in the machine or workplace master data. You can also use wildcards (placeholders *).

Group

This selection criterion refers to the group in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected group. You can also use wildcards.

Date from ... to ...

Fill in the fields from/to to narrow down the period to be evaluated

Shift / time

Restrict the defined period (date from/to). Select shifts or specify a time (from - until).

Report group

This selection criterion refers to the report groups. The application shows all machines or workplaces that are assigned to the selected report group.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected company. You can also use wildcards.

Responsibility area

This selection criterion refers to the responsibility area in the machine master data. Note: The user can only view those machines that are included in the responsibility areas that are assigned to the user.

Short name

This selection criterion refers to the short designation of machines in the master data. The application shows all machines or workplaces matching the entered string. You can also use wildcards.

Cost center

This selection criterion refers to the cost center stored in the machine or in the workplace master data. All machines or workplaces are displayed that are assigned to the selected cost center. You can also use wildcards.

Designation

This field refers to the name of machines and workplaces defined in the machine master data. Only those machines are displayed that are identical to the entered string. You can also use wildcards (placeholders *).

Counter type

In the detail application "Consumption figures", you can select the counters to be displayed using the counter type as defined within the counter configuration of the machine. To use this detail application, you require the relevant license.

Selection options

No selection: If you enable this option, you cannot select the other selection criteria (order and resource).

Order: You can use order parameters to restrict the data.

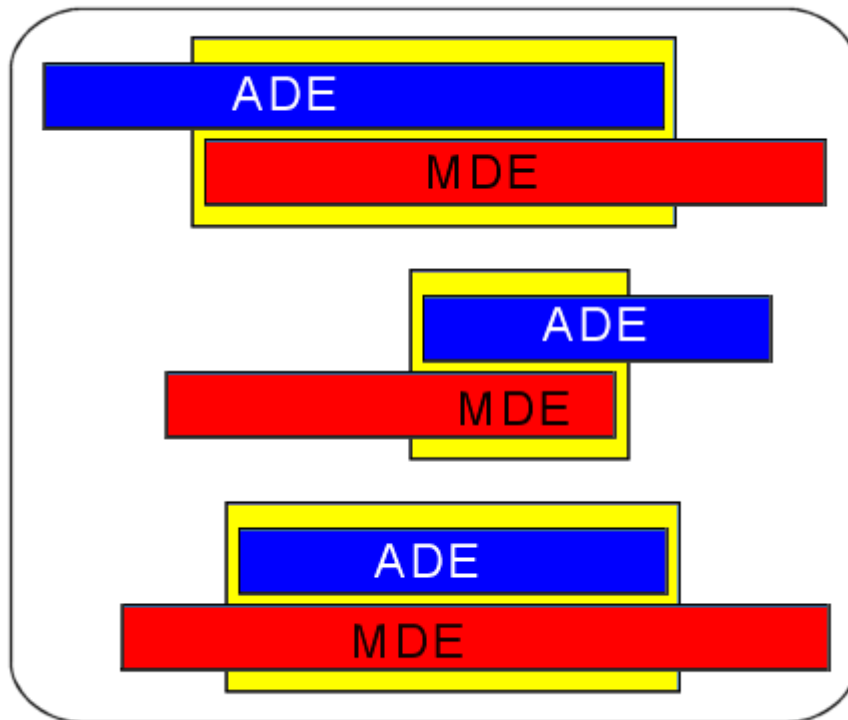
Resource: You can use resource parameters to restrict the data.

Order/Final article/Batch number

If you select these options, the application only includes completed ADE postings. If the order is currently still running on the machine, the system does not take into account the time period between the last logon and now. Therefore it is possible that differences appear between the machine evaluation and the order-related evaluation. The application only includes ADE postings that coincide with the evaluation period. If necessary, choose a selection period making sure that the required ADE postings coincide with this period. For this order-related evaluation, MPDV recommends selecting data via the "shift" option instead of the "time".

The order number, article/item or the batch (from the order header) may be used as selection criteria.

The illustration below shows an example of how ADE and MDE postings can overlap. The ADE postings take priority in this evaluation type. The yellow areas show the result of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If several orders are logged on to the machine simultaneously, this evaluation type (select one of the options *order*, *final article* or *batch number*) assigns the full machine time (yellow area) and quantities to every order. The fact that orders are run in parallel will not result in a proportionate calculation.

Long-term data



If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Resource/Resource type

Restrict the selected posting records integrated in the evaluation by selecting the logged in resources or resources of a specific resource type.

Detail application *Efficiency report*

The detail application *Efficiency report* includes workplace/machine-related performance data for a specific period of time and a specific number of workplaces. The result depends on the selection and therefore on the selection criteria available in the selection panel.

Category *Workplace*

The following workplace/machine-related master data are available:

- Workplace
- Short name
- Designation
- Group
- Cost center
- Company

Category *Primary quantity, Secondary quantity, Tertiary quantity, Basic quantity*

Workplace/machine-related quantities recorded in the corresponding quantity types

- Yield
- Scrap
- Rework
- Open quantity

or the relevant quantity units (if relevant in the customer system).

Category *Durations*

- Production = RPA11
- Downtime = RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10
- Total = production + downtime

Category *Cycles*

- Number of posted cycles

Category Key figures

- Rate of capacity utilization

The rate of capacity utilization is the quotient deriving from effective run time and machine working time.

Rate of capacity utilization = $100 / (RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10 + RPA11) * RPA11$

Formula definition: $rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11) * 100$

Use the formula **rcu** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Assignment utilization rate:

The assignment utilization rate represents the relationship deriving from the effective run time (main utilization time) and the adjusted machine operation time (i.e. without planned downtimes).

Assignment utilization rate = $100 / (\text{total of RPA 1 to RPA 11 minus RPA 6}) * RPA 11$

Formula definition: $rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa7+rpa8+rpa9+rpa10+rpa11) * 100$

Use the formula **ocu** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Efficiency (efficiency is only available when combined with the "lines" license)

Efficiency is the quotient derived from the effective run time and the general run time, meaning the sum total of the effective run time and any interruptions as a result of machine-related disturbances or unscheduled shutdowns. Any other disturbances, e.g. organization-related disturbances, are not taken into account here.

Efficiency = $100 / (RPA02 + RPA05 + RPA11) * RPA11$

- Techn. efficiency

The reference value for the technical efficiency is the sum total of the effective run time and interruptions because of technical (machine-related) disturbances. Other interruptions (e.g. of a organizational nature) are not considered: Techn. efficiency = $100 / (RPA02 + RPA11) * RPA11$

Formula definition: $rpa11 / (rpa2 + rpa11) * 100$

Use the formula **tec_ef** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- $\text{Rate} = 100 / ((\text{yield (primary quantity)} + \text{scrap (primary quantity)}) * \text{rework (primary quantity)} + \text{open quantity (primary quantity)}) * \text{yield (primary quantity)}$

Formula definition: $\text{yield.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}) * 100$

Use the formula **yie_ra** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- $\text{Scrap rate} = 100 / ((\text{yield (primary quantity)} + \text{scrap (primary quantity)} + \text{rework (primary quantity)} + \text{open quantity (primary quantity)}) * \text{scrap (primary quantity)})$

Formula definition: $\text{scrap.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}) * 100$

Use the formula **scr_ra** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

Note:

If you select the "time" option, the application only includes those MDE log records that are completely within the selected period (start and end must be within the selected period), that extend into the selected period (start or end is within the time frame) or that cover the complete selection period.

If you select the current shift where no MDE log records have been posted yet, the quantities produced within the selected period of time are calculated proportionally.



Example: At 10:00 am, you select an evaluation of the current shift from 8:00 to 9:00 am and the machine has the status production since 8:00 am. To calculate the recorded quantity in the selected period of time, the quantity produced since 8:00 (until 10:00) (1200 pieces) is divided according to the following formula:

$((\text{produced yield} / \text{complete duration of the current status in seconds}) * \text{evaluation duration})$

In this example it is: $(1200 / 7200) * 3600 = 600$ pieces. The efficiency report will show a produced quantity of 600 pieces.

This proportionate calculation can have the effect that evaluated quantities can change within the current shift even if the selection parameters have not changed.

Detail application *Quantitative activities (machine-related)*

This detail application generates a graphic presentation of the quantities for the workplaces selected in the tabular detail application. Here, a differentiation is made by yield, scrap, rework and open quantity.



This detail application and the detail application *Quantity rate (group-related)* are docked one behind the other.

Detail application *Quantity rate (group-related)*

This chart shows, based on the selected entries in the table, the relationship between the quantities - yield, scrap, rework and open quantities. The information is displayed by group/ added total by workplace group.



This detail application and the detail application *Quantitative activities (machine-related)* are docked one behind the other.

Detail application *Durations*

This detail application generates a graphic illustration of the selected data from the efficiency report detail application. The durations are shown, broken down by RPA accounts (RPA01... RPA12).

You can hover the mouse over the graphic to display the value of the area where the mouse is. You can switch between displaying the value in percent or in duration.

Detail application *Consumption figures*

Shows the master data and counter values of the machine counters configured for the machines. By selecting the counter type, the data displayed can be limited to e.g. consumption figures or yield counters, or similar.



This detail application is only available if the system is configured accordingly and the relevant licenses are available.

3 Performance profile

Overview

Menu	Production facility management → Key performance indicators → Performance profile
Transaction code	effpf
Function authorization	effpf

The *Performance profile* application provides a tabular and graphic presentation of the production performance interrelationships. The collected performance data is compressed to days and shifts. This way, the application can provide indispensable production KPIs for all persons in a position of responsibility. This application provides a clear overview of all quantities and durations that are necessary to reliably assess the production status.

Selection criteria

The application provides the following selection criteria:

Workplace

This selection criterion refers to the workplace in the machine or workplace master data. You can also use wildcards (placeholders *).

Group

This selection criterion refers to the group in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected group. You can also use wildcards.

Date from ... to ...

Fill in the fields from/to to restrict the period to be evaluated

Shift / time

Restrict the defined period (date from/to). Select shifts or specify a time (from - until).

Report group

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected report group.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected company. You can also use wildcards.

Responsibility area

This selection criterion refers to the responsibility area in the machine master data. Note: The user can only view those machines that are included in the responsibility areas that are assigned to the user.

Short name

This selection criterion refers to the short designation of machines in the master data. The application shows all machines or workplaces matching the entered string. You can also use wildcards.

Cost center

This selection criterion refers to the cost center stored in the machine or in the workplace master data. All machines or workplaces are displayed that are assigned to the selected cost center. You can also use wildcards.

Name (designation)

This field refers to the name of machines and workplaces defined in the machine master data. The application only shows those machines matching the entered string. You can also use wildcards (placeholders *).

Selection options

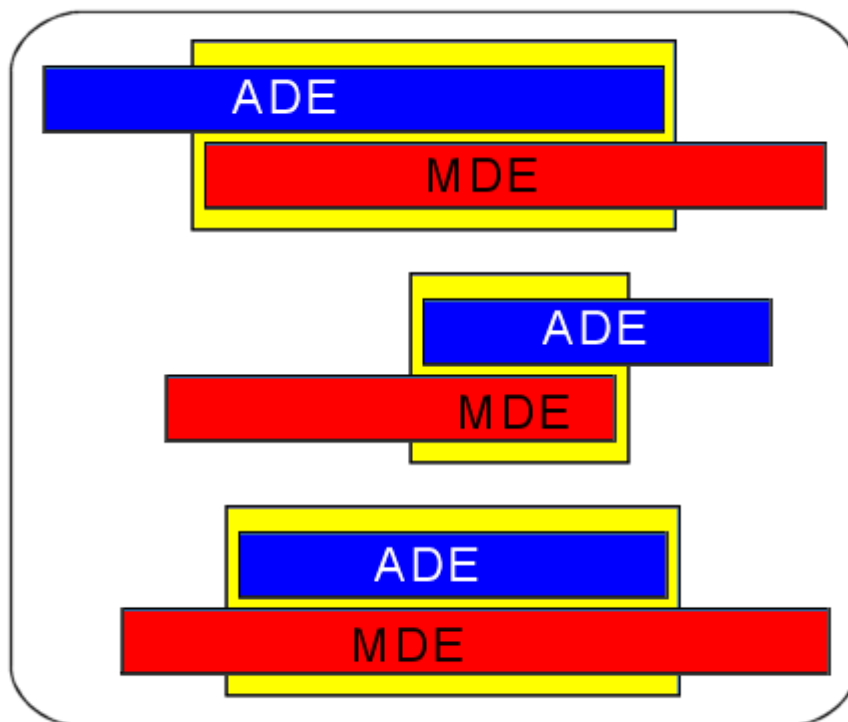
- No selection:* If you enable this option, you cannot select the other selection criteria (order and resource).
- Order:* You can use order parameters to restrict the data.
- Resource:* You can use resource parameters to restrict the data.

Order/final article/batch number

If you select these options, the application only includes completed ADE postings. If the order is currently still running on the machine, the system does not take into account the time period between the last logon and now. Therefore it is possible that differences appear between the machine evaluation and the order-related evaluation. The application only includes ADE postings that coincide with the evaluation period. If necessary, choose a selection period making sure that the required ADE postings coincide with this period. For this order-related evaluation, MPDV recommends selecting data via the "shift" option instead of the "time".

The order number, article/item or the batch (from the order header) may be used as selection criteria.

The illustration below shows an example of how ADE and MDE postings can overlap. The ADE postings take priority in this evaluation type. The yellow areas show the result of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If several orders are logged on to the machine simultaneously, this evaluation type (select one of the options *order*, *final article* or *batch number*) assigns the full machine time (yellow area) and quantities to every order. The fact that orders are run in parallel will not result in a proportionate calculation.

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If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Resource/resource type

Restrict the selected posting records integrated in the evaluation by selecting the logged in resources or resources of a specific resource type.

Group result

Use this selection criterion to group the results according to the following parameters:

- Year
- Month
- Calendar week
- Date
- Shift date and shift

Performance profile detail application

The performance profile of machine-related performance data is presented for a specific period of time and a certain number of workplaces. The result depends on the selection made and therefore on the selection criteria available in the selection panel.

Date category

Depending on the selection of the option "Group results" in the selection panel, the columns are completed as shown below (the columns that are not listed for the respective selection remain empty):

- Select shift date and shift: the application shows the year, calendar week number, month, shift date, shift.
- Select date: the application shows the year, calendar week number, month, shift date.
- Select week: the application shows the year, calendar week number.
- Select month: the application shows the year, month.
- Select year: the application shows the year.

Primary quantity, secondary quantity, tertiary quantity, basic quantity category

Workplace/ machine-related quantities recorded in the corresponding quantity types

- Yield
- Scrap
- Rework
- Open quantity

or the relevant quantity units (if relevant in the customer system).

Durations category

- Production = RPA11
- Downtime = RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10
- Total = production + downtime

Key figures category

- Rate of capacity utilization

The rate of capacity utilization is the quotient deriving from effective run time and machine working time.

Rate of capacity utilization = $100 / (RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10 + RPA11) * RPA11$

Formula definition: $rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11) * 100$

Use the formula **rcu** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Assignment utilization rate:

The assignment utilization rate represents the relationship deriving from the effective run time (main utilization time) and the adjusted machine operation time (i.e. without planned downtimes).

Assignment utilization rate = $100 / (\text{total of RPA 1 to RPA 11 minus RPA 6}) * RPA 11$

Formula definition: $rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa7+rpa8+rpa9+rpa10+rpa11) * 100$

Use the formula **ocu** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Efficiency (efficiency is only available when combined with the "lines" license)

Efficiency is the quotient derived from the effective run time and the general run time, meaning the sum total of the effective run time and any interruptions as a result of machine-related disturbances or unscheduled shutdowns. Any other disturbances, e.g. organization-related disturbances, are not taken into account here.

Efficiency = $100 / (RPA02 + RPA05 + RPA11) * RPA11$

- Techn. efficiency

The reference value for the technical efficiency is the sum total of the effective run time and interruptions because of technical (machine-related) disturbances. Other interruptions (e.g. of a organizational nature) are not considered: Techn. efficiency = $100 / (RPA02 + RPA11) * RPA11$

Formula definition: $rpa11 / (rpa2 + rpa11) * 100$

Use the formula **tec_ef** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Rate = $100 / ((\text{yield (primary quantity)} + \text{scrap (primary quantity)} + \text{rework quantity (primary quantity)} + \text{open quantity (primary quantity)}) * \text{yield (primary quantity)})$.

Formula definition: $\text{yield.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}) * 100$

Use the formula **yie_ra** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Scrap rate = $100 / ((\text{yield (primary quantity)} + \text{scrap (primary quantity)}) * \text{scrap (primary quantity)})$

Formula definition: $\text{scrap.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}) * 100$

Use the formula **scr_ra** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

Quantities detail application

This chart shows, based on the selected entries in the table, the relationship between the quantities.

Depending on the option "group result", that you have enabled in the selection range, the values are displayed per

- Shift/shift date (select shift date and shift)

1	2	3
10.01.2011	10.01.2011	10.01.2011
- Shift date (select shift date)

09.01.2011	10.01.2011	11.01.2011
------------	------------	------------
- Calendar week (select calendar week)

2011	2011	2011
Woche: 1	Woche: 2	Woche: 3
- Month (select month)

2011/1	2011/2
--------	--------
- Year (select year)

2010	2011
------	------

You can use the "Displayed series" combo box to select which quantities you would like to have displayed (e.g. only yield and scrap).

Durations detail application

The durations detail application shows, based on the selected entries in the table, the relationship between the production duration and downtime. Depending on the option "group result", that you have enabled in the selection range, the values are displayed per

- Shift/shift date (select shift date and shift)

- Shift date (select shift date)
- Calendar week (select calendar week)
- Month (select month)
- Year (select year)



This detail application is docked right behind the key figures detail application.

Key figures detail application

The key figures detail application shows, based on the selected entries in the table, the efficiency or the technical efficiency, assignment utilization rate, rate, scrap rate or the rate of capacity utilization in graphic form.

Select the relevant key figure from the combo box. If no key figure is selected, the rate of capacity utilization is shown.

Depending on the option "group result", that you have enabled in the selection range, the key figures are displayed per

- Shift/shift date (select shift date and shift)
- Shift date (select shift date)
- Calendar week (select calendar week)
- Month (select month)
- Year (select year)

The key figure in the label (if activated) is displayed to the second decimal place.



This detail application is docked right behind the durations detail application.

4 OEE report

Overview

Menu	Production facility/resource management --> Key performance indicators --> OEE report
Transaction code	oeerp
Function authorization	oeerp

The report concerns machine-related OEE performance data for a certain period of time and number of workstations. The result depends on the selections made and therefore on the selection criteria available in the selection panel. This report lists meaningful data in tabular form and provides vital information in graphic layouts. This makes it a useful tool for all users.

The application uses machine-related postings to calculate KPIs.

Selection criteria

The application provides the following selection criteria:

Workplace

The selection criterion *workplace* refers to the workplace in the machine or workplace master data. You can also use wildcards (placeholders *).

Group

The selection criterion *group* refers to the group in the machine or workplace master data. The application shows all machines and/or workplaces that are assigned to the selected group. You can also use wildcards (placeholders *).

Date from ... to ...

The selection criterion *date from ... to ...* restricts the period you want to evaluate.

Shift / time

Use the selection criterion *shift and/or time* to further restrict the specified period (date from ... to ...). To do so, select a shift or specify a time (from ... until...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

Report group

The selection criterion *Report group* refers to the report groups. The application shows all machines or workplaces that are assigned to the selected report group.

Company

The selection criterion *company* refers to the company defined in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected company. You can also use wildcards (placeholders *).

Responsibility area

The selection criterion *responsibility area* refers to the responsibility area in the machine master data. Note that you can only view those machines you are authorized for (responsibility area).

Short name

The selection criterion *short name* refers to the short name of machines in the master data. The application shows all machines or workplaces matching the entered string. You can also use wildcards (placeholders *).

Cost center

The selection criterion *cost center* refers to the cost center stored in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected cost center. You can also use wildcards (placeholders *).

Name (designation/description)

The field *name* refers to the name of machines and workplaces defined in the machine master data. This field only shows the machines matching the entered string. You can also use wildcards (placeholders *).

Selection options

No selection: If you choose this field, you cannot select the other selection criteria (order and resource).

Order: Use order parameters to restrict the data.

Resource: Use resource parameters to restrict the data.

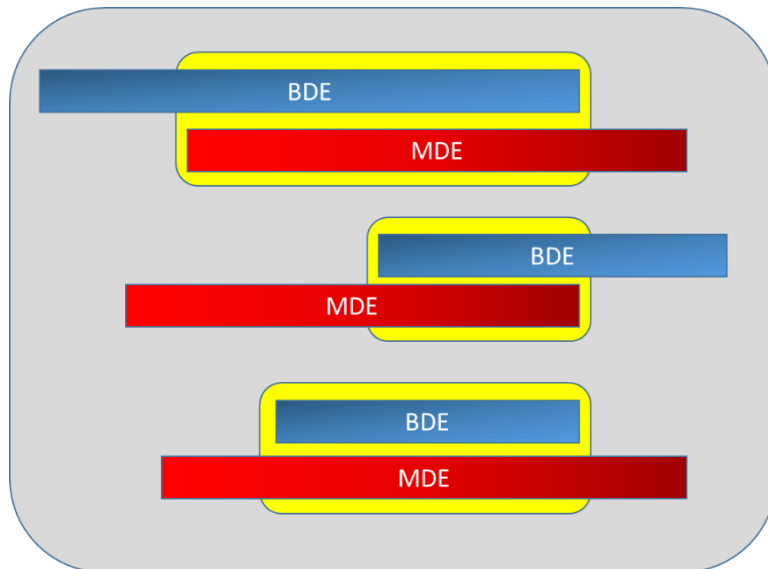


If you check one of the options "order" or "resource", the application calculates the KPIs (key performance indicators) based on the machine-related postings that occurred while an order and/or resource was logged on. The application does not use the data posted onto the order and/or resource to calculate the KPIs. The application uses machine-related postings to calculate KPIs.

Order/Final article/Batch number

If you select the evaluation option *order*, the application identifies the BDE postings starting within the selection period and matching the entered selection criteria *order*, *finished article*, *batch number* (from the order header). If necessary, choose a selection period making sure that the required BDE postings actually coincide with this period.

The application compares these BDE postings and their periods with the machine-related postings. The illustration below shows an example of how BDE and MDE postings can overlap. The yellow areas show the result of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If several orders are logged on to the machine simultaneously, this evaluation type (select one of the options *order*, *final article* or *batch number*) assigns the full machine time (yellow area) and quantities to every order. The fact that orders are run in parallel will not result in a proportionate calculation.



The application only integrates completed BDE postings. If the operation is still logged on to the machine when you perform the evaluation, the time period between the last logon and now is not taken into account. Therefore, differences might appear between the machine evaluation and the order-related evaluation. For this evaluation type (select one of the options *order*, *finished article*, *batch number*), MPDV recommends selecting data via the "shift" option instead of the "time" option.

Resource/Resource type

If you select the evaluation option *resource*, the application restricts the selected posting records integrated in the evaluation based on the logged in resources and/or the resources of a specific resource type. Then the application compares these posting records with the machine-related postings as described above.

OEE report detail application

The detail application *OEE report* refers to machine-related OEE performance data for a certain period of time and number of workplaces.

Master data category

The following workplace/machine-related master data are available:

- Workplace
- Short name
- Name (designation/description)
- Group
- Cost center
- Company

Category durations

In addition to master data and the below-mentioned KPIs, the application shows the following data in tabular form:

- Planned operating time
- Machine runtime
- Actual utilization
- Yield utilization

Refer to the [document dealing with OEE calculation](#) for further information on the calculation options.

Category Key figures

The OEE is calculated as follows:

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

Refer to the [document dealing with OEE calculation](#) for further information on the calculation options.

Total line detailed application OEE report

The total line for the categories *durations* and *cycles* totals the values of these categories.

The total line of the category KPIs, on the other hand, does not provide a total but an average and is calculated differently from the standard average (sum of all values divided by number).

Master data			Durations				Key figures			
Workplace	Short name	Group ▲	Planned oper. time	Machine run time	Actual utilization	Yield utilization	OEE	Availability	Perform...	Quality
50611	ARB-320S	INJ.MOL...	86:45:47	77:33:41	66:12:59	71:44:24	0,76	0,89	0,92	0,92
50612	ARB-320S	INJ.MOL...	529:28:10	58:59:33	55:46:19	58:30:41	0,11	0,11	0,95	0,99
IDX01	INDEX01	NC	274:55:01	232:56:12	17:49:59	192:19:48	0,06	0,85	0,09	0,83
IDX02	INDEX02	NC	137:17:25	133:36:15	124:49:19	127:59:51	0,91	0,97	0,98	0,96
MOR12	MORISEI2	NC	274:25:13	245:41:25	33:16:13	235:24:56	0,12	0,90	0,14	0,96
NA42	NAKAMU42	NC	66:06:30	56:31:12	48:25:58	54:27:43	0,73	0,85	0,89	0,96
TR1	TRUMPF	PUNCHING	346:57:35	345:44:08	129:33:59	342:46:30	0,37	1,00	0,38	0,99
TRU600	TRUMATIC	PUNCHING	274:25:32	62:23:30	57:46:19	60:20:33	0,21	0,23	0,96	0,97
TRU601	TRUMAT02	PUNCHING	62:24:58	52:55:43	46:14:30	50:02:58	0,74	0,85	0,92	0,95
BI40	BIHLER40	PUNCHI...	75:13:00	34:28:22	22:49:18	24:26:04	0,30	0,46	0,93	0,71
BI41	BIHLER41	PUNCHI...	273:46:21	269:45:56	56:58:13	262:21:25	0,21	0,99	0,22	0,97
60610	ENG21	SGM2	64:20:26	58:48:22	55:10:07	56:50:48	0,86	0,91	0,97	0,97
			2806:17:41	1947:45:50	990:30:06	1841:56:53	0,36	0,69	0,53	0,97

Example:

MDE posting records

Machine	Strokes	Yield	Scrap	Target cycle	Target cycle per 1	RPA	Duration
50611	4163	24180	360	4468	4.468	11	20188
50611	150	0	584	4968	4.968	3	789
50611	1613	0	0	4468	4.468	11	7823
Total		24180	944				
50614	953	22500	4980	6984	6.984	11	7171
50614	514	0	2044	6984	6.984	3	3863
50614	2360	0	0	7500	7.500	11	17766
		22500	7024				
Total		46680	7968				

OEE calculation Machine 50611:

Key Performance Indicator	Formula	Calculation	Result
Quality	$\frac{\text{Yield}}{\text{Yield} + \text{Scrap} + \text{Open quantity} + \text{Rework}}$	24180 / (24180+944)	0.96
Availability	$\frac{\sum RPA_{11}}{\sum (RPA_{1-11})}$	(20188+7823)/(20188+789+7823)	0.97
Performance	Performance = $\frac{\text{Target cycle}}{\text{Actual cycle}}$ Actual cycle = $\frac{\sum RPA_{11}}{\sum \text{Stroke}}$ The individual performance values are weighted in order to show a consolidated view of the evaluations. Weighting (Performance_{Product}) is based on production time (RPA11). Performance_{Product} = Performance * RPA 11 The values of the summarized performance (Performance_{Product}) are divided after the summation by the sum of the production time (RPA11)	4.468/(7823/1613) = 0.921 0.921 * 7823 = <u>7207.748</u> 4.468/(20188/4163) = 0.921 0.921 * 20188 = <u>18600.284</u> (7207.748 + 18600.284) / (7823+20188) = 0.921	0.92

	to get the corresponding performance. $\text{Performance} = \frac{\sum \text{Performance}_{\text{Product}}}{\sum \text{RPA}_{11}}$		
OEE	Quality* Availability * Performance	0.96 * 0.97 * 0.92	0.86

OEE calculation Machine 50614:

Key Performance Indicator	Formula	Calculation	Result
Quality	$\frac{\text{Yield}}{\text{Yield} + \text{Scrap} + \text{Open quantity} + \text{Rework}}$	22500 / (22500 + 7024)	0.76
Availability	$\frac{\sum \text{RPA}_{11}}{\sum (\text{RPA}_{1-11})}$	(7171+17766)/ (7171+3863+17766)	0.87
Performance	$\text{Performance} = \frac{\text{Target cycle}}{\text{Actual cycle}}$ $\text{Actual cycle} = \frac{\sum \text{RPA}_{11}}{\sum \text{Stroke}}$ The individual performance values are weighted in order to show a consolidated view of the evaluations. Weighting (Performance_{Product}) is based on production time (RPA11). $\text{Performance}_{\text{Product}} = \text{Performance} * \text{RPA}_{11}$ The values of the summarized performance (Performance_{Product}) are divided after the summation by the sum of the production time (RPA11) to get the corresponding performance. $\text{Performance} = \frac{\sum \text{Performance}_{\text{Product}}}{\sum \text{Stroke}_{11}}$	6.984/(7171/953) = 0.9281 0.9281 * 7171 = <u>6655.752</u> 7.500/(17766/2360) = 0.9962 0.9962 * 17766 = <u>17700</u> (6655.752 + 17700) / (7171+17766) = 0.98	0.98
OEE	Quality* Availability * Performance	0.96 * 0.97 * 0.92	0.86

OEE calculation of the total line

Key Performance Indicator	Formula	Calculation	Result
Quality	$\frac{\text{Yield}}{\text{Yield} + \text{Scrap} + \text{Open quantity} + \text{Rework}}$	(22500+24180) / (22500 + 7024 + 24180 + 944)	0.85
Availability	$\frac{\sum \text{RPA}_{11}}{\sum (\text{RPA}_{1-11})}$	(7171+17766+20188+7823)/ (7171+3863+17766+20188+789+7823)	0.92

Performance	$Performance = \frac{\text{Target cycle}}{\text{Actual cycle}}$ $\text{Actual cycle} = \frac{\sum RPA\ 11}{\sum \text{Stoke}}$ The individual performance values are weighted in order to show a consolidated view of the evaluations. Weighting ($Performance_{Product}$) is based on production time (RPA11). $Performance_{Product} = \text{Performance RPA} \cdot$ The values of the summarized performance ($Performance_{Product}$) are divided after the summation by the sum of the production time (RPA11) to get the corresponding performance. $Performance = \frac{\sum Performance_{Product}}{\sum RPA\ 11}$	$6.984/(7171/953) = 0.9281$ $0.9281 \cdot 7171 = \underline{6655.752}$	0.95
		$7.500/(17766/2360) = 0.9962$ $0.9962 \cdot 17766 = \underline{17700}$	
		$4.468/(7823/1613) = 0.921$ $0.921 \cdot 7823 = \underline{7207.748}$	
		$4.468/(20188/4163) = 0.921$ $0.921 \cdot 20188 = \underline{18600.284}$	
		$(6655.752 + 17700 + 7207.748 + 18600.284) / (7171 + 17766 + 7823 + 20188) = 0.947$	
OEE	Quality* Availability * Performance	$0.85 \cdot 0.92 \cdot 0.95$	0.74

OEE display in the MOC for the above example:

OEE report											
Drag a column here to group by that column											
Master data			Durations				Key figures				
Workplace	Short name	Group	Planned oper. time	Machine run time	Actual utilization	Yield utilization	OEE	Availability	Perform...	Quality	
50611	ARB-320S	SPRGUSS1	8:00:00	7:46:51	6:53:57	7:29:18	0,86	0,97	0,92	0,96	
50614	ARB-270C	SPRGUSS1	8:00:00	6:55:37	5:09:21	5:16:44	0,64	0,87	0,98	0,76	
			16:00:00	14:42:28	12:03:18	12:46:02	0,74	0,92	0,95	0,85	

Machine-related KPIs detail application

The diagram *machine-related KPIs* shows the following KPIs in a bar chart based on the machines selected in the table.

- OEE
- Availability
- Performance
- Quality

Group-related KPIs detail application

The diagram *group-related KPIs* shows the following group-related KPIs in a bar chart based on the machines selected in the table.

- OEE
 - Availability
 - PerformanceQuality
- ### **Basic data detail application**

The detail application *Basic data* displays basic data in graphic format:

- Planned operating time
- Machine run time
- Actual utilization
- Yield utilization

This application shows basic data for each workplace selected in the tabular detail application.

Values are *durations*, X-axis labeling is carried out automatically.